TerraAlert v3: Model Evaluation Report

Generated 2026-11-13 by ai-module/v3/research_evaluation_v3.py

Abstract

We evaluate an XGBoost model that rates landslide risk from rainfall, terrain, vegetation and soil inputs. On 1,610 rows (552 landslides) from Himachal Pradesh, Uttarakhand and Kashmir, using 5-fold spatial cross-validation with ~100 km blocks held out, the tuned model reaches ROC-AUC 0.613 [0.587, 0.639] versus 0.572 [0.547, 0.597] for the untuned model and 0.489 [0.468, 0.514] for the earlier v2. At the app's alert level it catches 70% of landslides while flagging 53% of safe rows. Strict nested cross-validation gives ROC-AUC 0.609. The model is a risk indicator, not a precise slope-by-day predictor.

1. Data

Case-control design: each catalogued landslide (label 1) is paired with a nearby slope on the same day (5-30 km away, elevation within 300 m) and the same location on the same date in a year without a landslide (both label 0). Inputs come from the same feature code the live app uses (NASA IMERG V07 rain, SRTM terrain, MODIS vegetation, OpenLandMap soil).

Row type	Rows
Landslide	552
Same place, normal year	552
Nearby slope, same day	506
Spatial blocks (CV groups)	27

2. Evaluation protocol

All scores are out-of-fold from 5-fold StratifiedGroupKFold, where a group is a 1 x 1 degree block, so test areas are never seen in training. 95% confidence intervals come from a cluster bootstrap (1000 resamples of whole blocks). The tuned settings were chosen on all rows, so the out-of-fold figures here are slightly optimistic. The strictly separated nested-CV result is reported alongside. Operating points: **alert** = the app's MODERATE level (chosen to catch 70% of landslides for v3 tuned, and score >= 40 for the old models as deployed); **matched** = each model's cut-off for 70% recall, which compares models at equal catch rates.

3. Results

Metric	v3 tuned	v3 original	v2
ROC-AUC [95% CI]	0.613 [0.587, 0.639]	0.572 [0.547, 0.597]	0.489 [0.468, 0.514]
PR-AUC [95% CI]	0.433 [0.401, 0.468]	0.412 [0.390, 0.443]	0.327 [0.313, 0.346]
Brier score (lower = better)	0.238	0.255	0.406
At alert level as deployed			
Cut-off	0.47	0.40	0.40
Recall (caught)	70%	41%	60%
Precision	41%	41%	34%
False alarm rate	53%	31%	59%
Specificity	47%	69%	41%
NPV	75%	69%	66%
Accuracy	55%	59%	47%
Balanced accuracy	58%	55%	50%
F1	0.51	0.41	0.44
F2 (recall-weighted)	0.61	0.41	0.52

Metric	v3 tuned	v3 original	v2
MCC	0.16	0.10	0.00
At matched 70% recall			
Cut-off	0.47	0.18	0.20
Recall (caught)	70%	70%	70%
Precision	41%	38%	35%
False alarm rate	53%	61%	69%
Specificity	47%	39%	31%
NPV	75%	71%	66%
Accuracy	55%	50%	44%
Balanced accuracy	58%	55%	50%
F1	0.51	0.49	0.46
F2 (recall-weighted)	0.61	0.60	0.58
MCC	0.16	0.09	0.01

Paired cluster bootstrap, v3 tuned minus v3 original ROC-AUC: +0.041 [95% CI +0.015, +0.063]. Nested CV (strict): tuned 0.609 vs original 0.577; overfitting gap 0.43 -> 0.07.

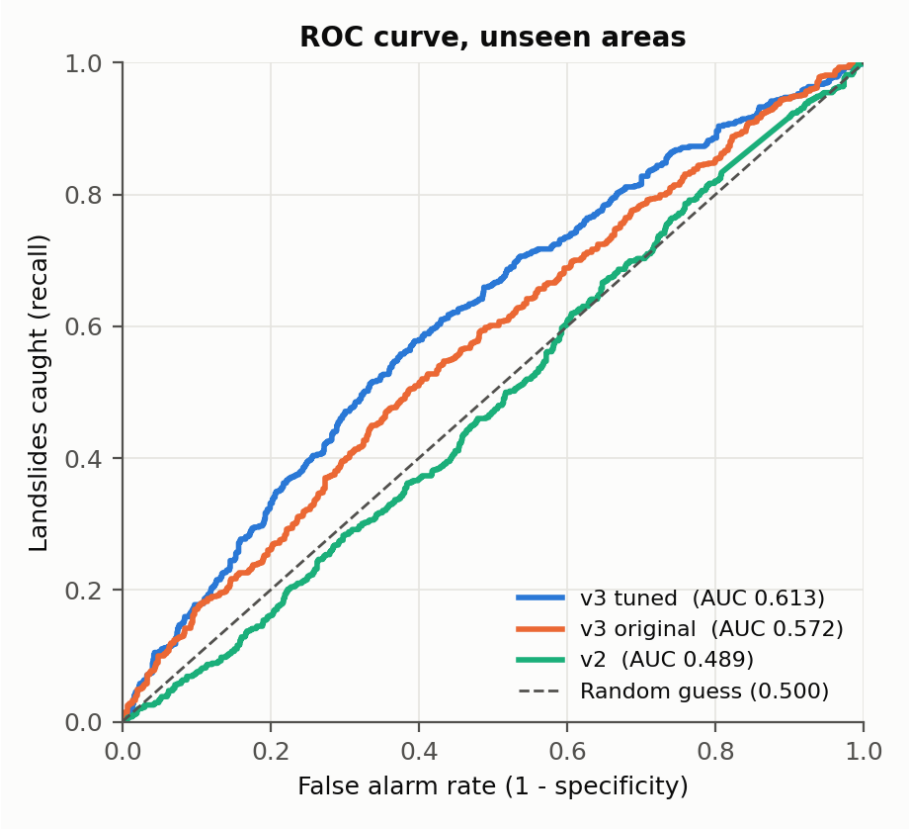


Figure 1. ROC curves on unseen areas. The dashed line is random guessing.

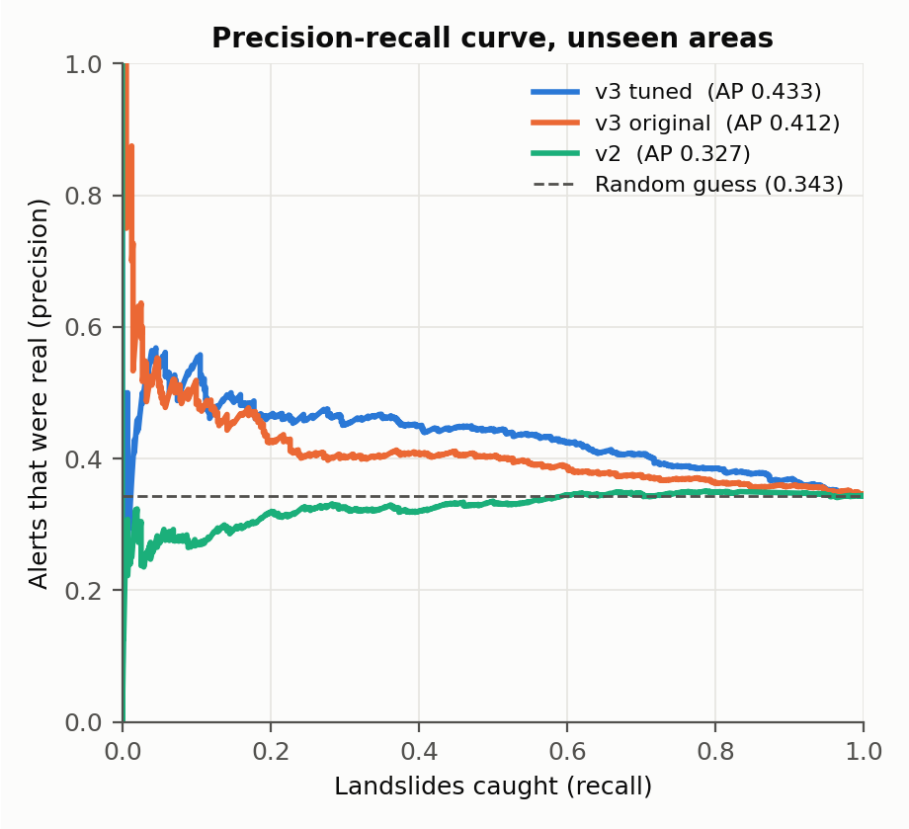


Figure 2. Precision-recall curves. The dashed line is the share of landslides in the data.

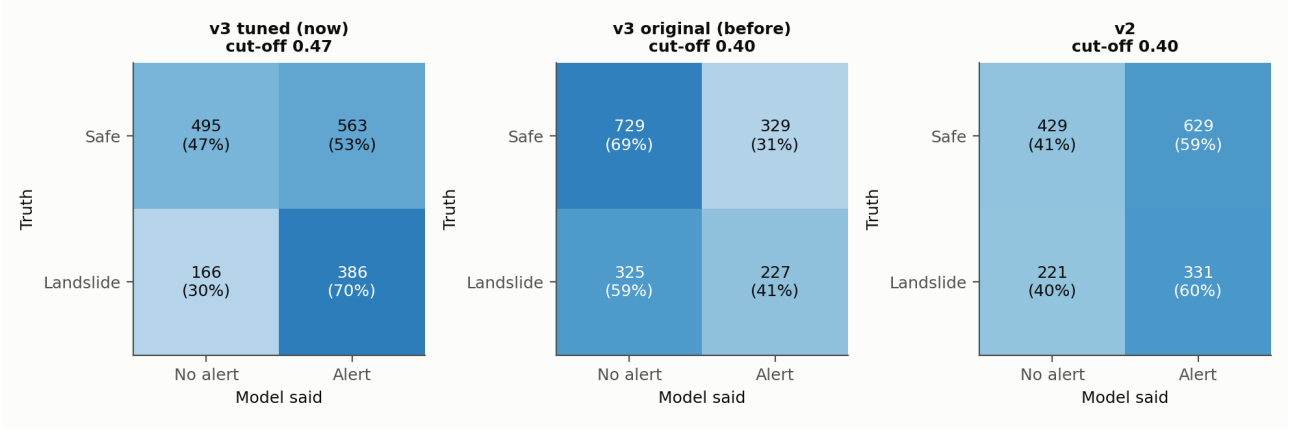


Figure 3. Confusion matrices at the alert level, as deployed. Cells show counts and the share of each true class.

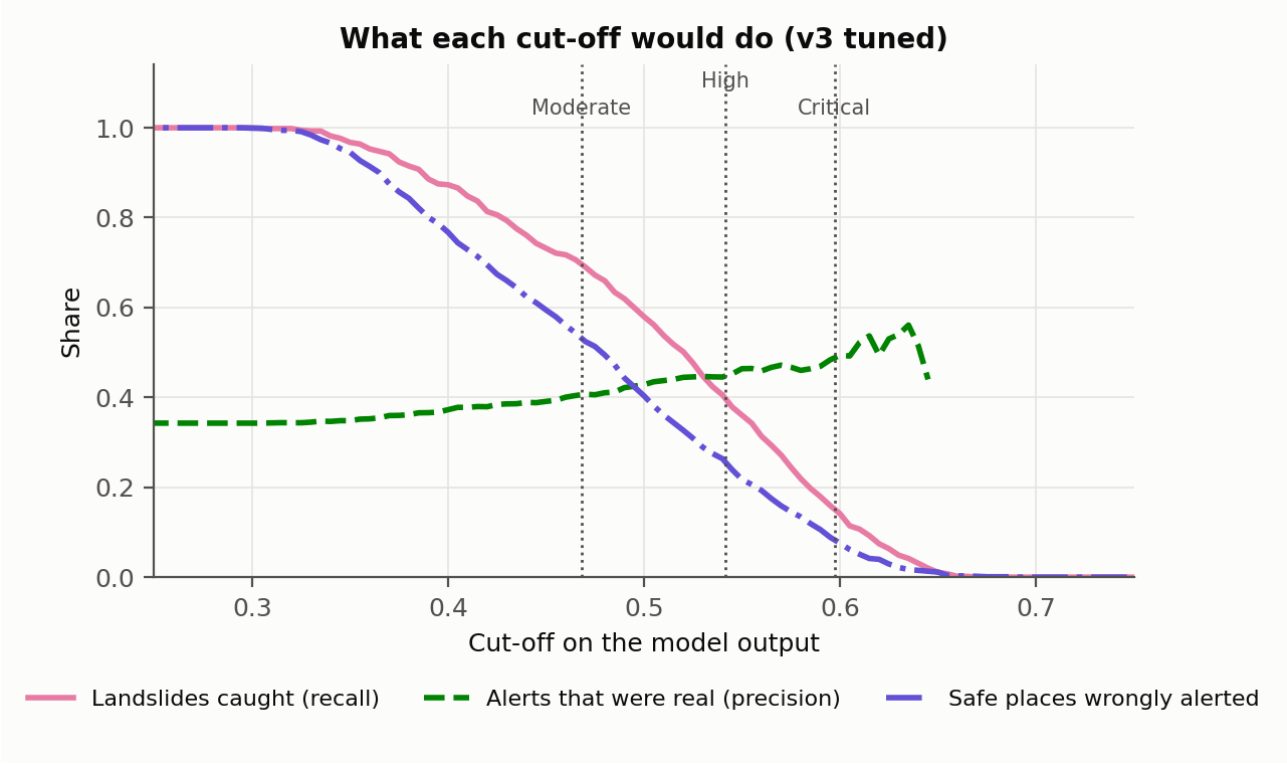


Figure 4. Recall, precision and false-alarm rate for every possible cut-off (v3 tuned). Dotted lines mark the app's level boundaries.

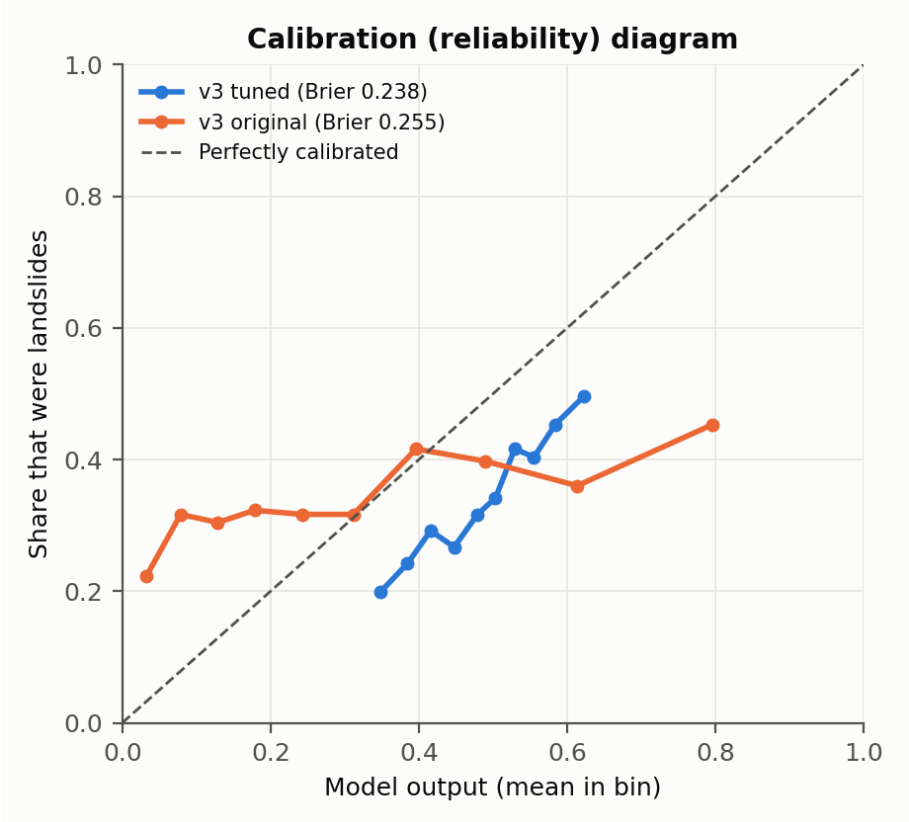


Figure 5. Calibration. Training used 1 landslide per 2 safe rows, so raw outputs are not real-world probabilities; the app therefore maps outputs to levels by recall instead.

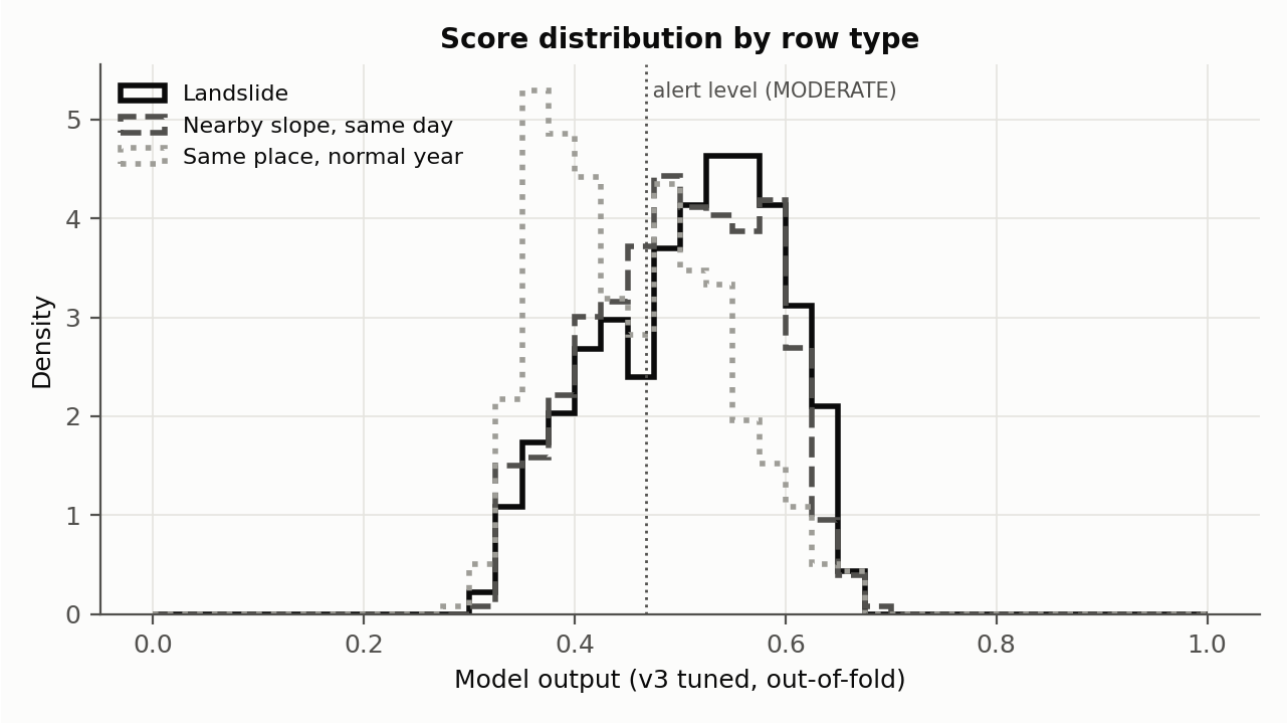


Figure 6. Out-of-fold score distribution by row type. Landslides and normal-year rows separate; nearby same-day slopes overlap heavily with landslides.

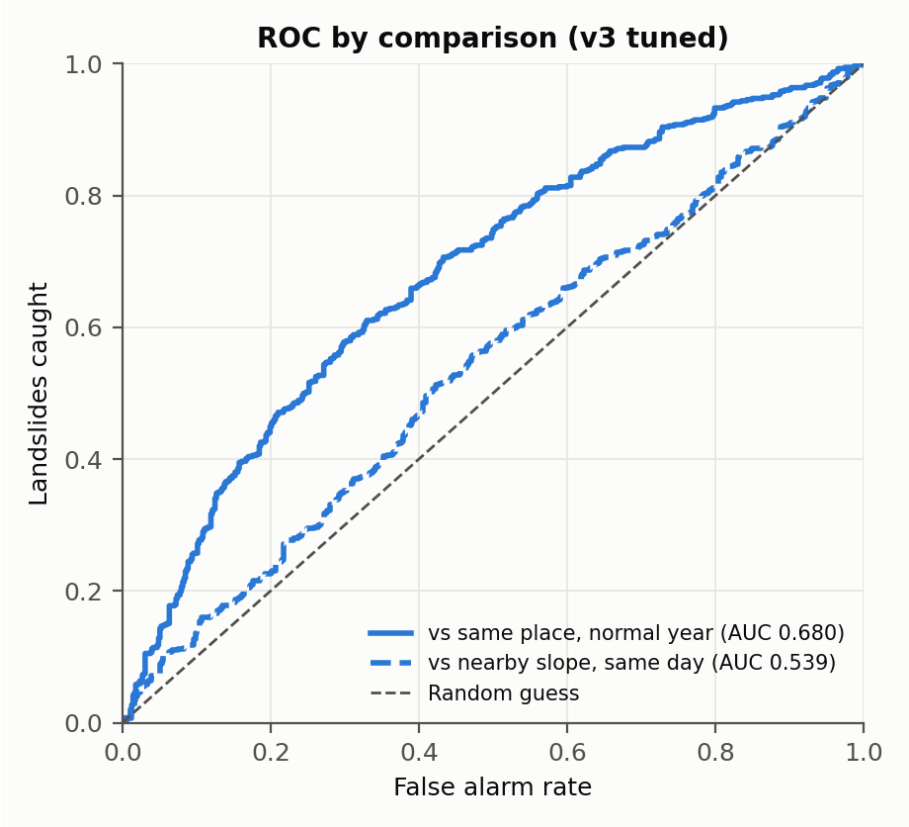


Figure 7. ROC by comparison type: the model separates landslide days from normal days at the same place much better than neighbouring slopes on the same day.

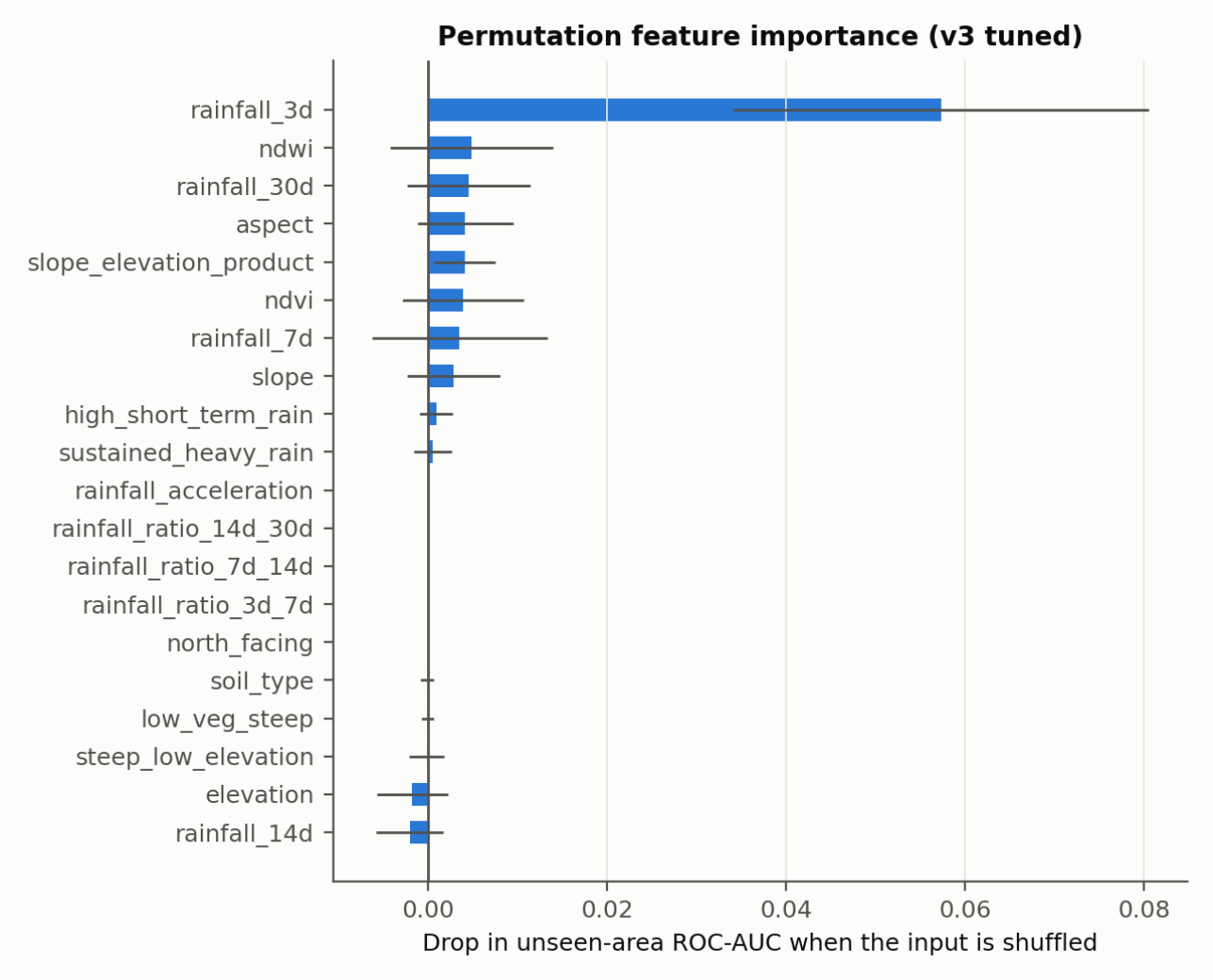


Figure 8. Permutation importance on held-out folds (mean and SD over 5 folds x 5 shuffles). Inputs held constant in training show zero by design.

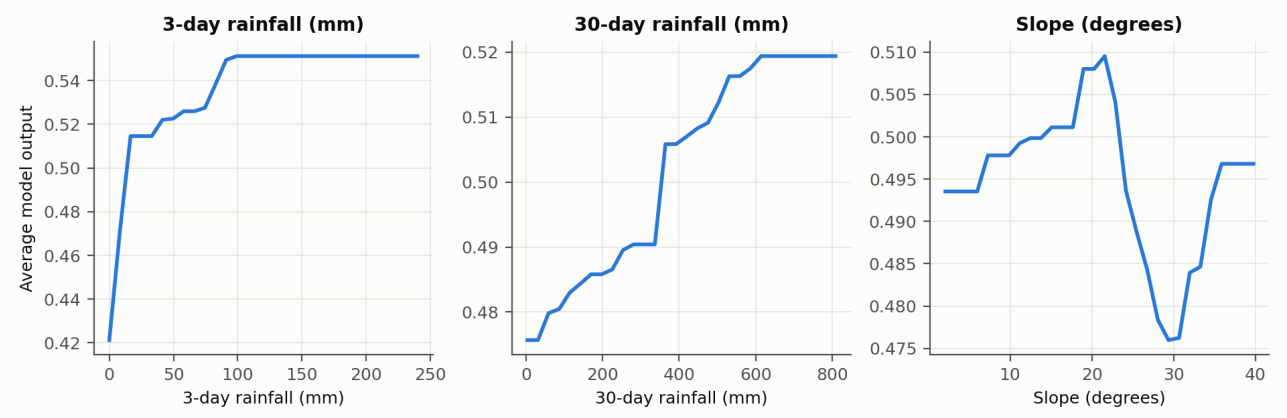


Figure 9. Partial dependence: rain can only raise the output (monotone constraint), while slope has its own learned shape.

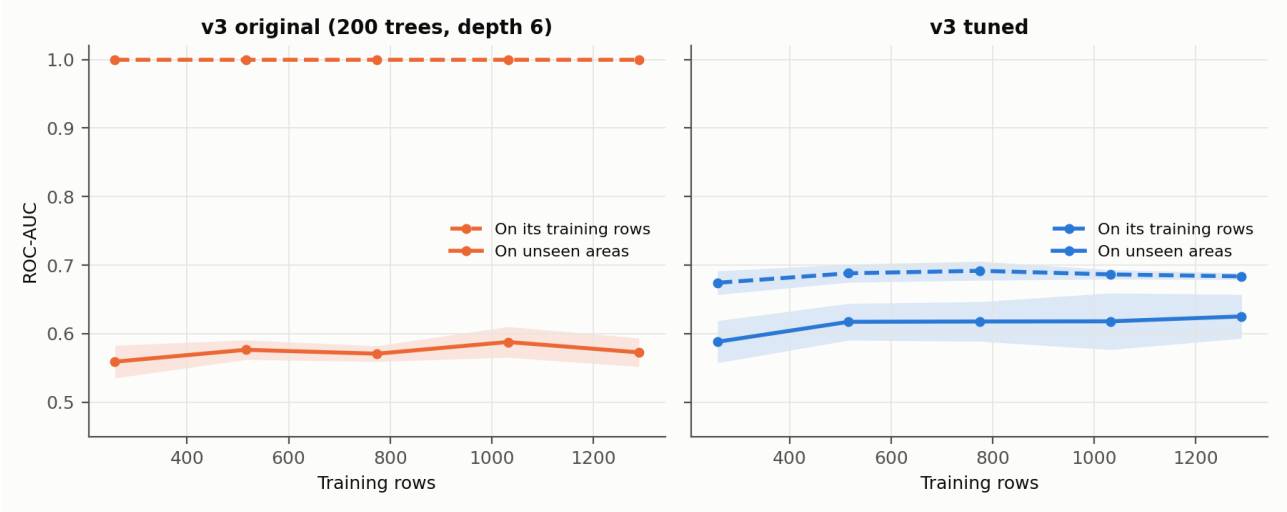


Figure 10. Learning curves (mean +/- SD over folds). The original model reaches about 1.0 on its training rows (memorisation); the tuned model keeps training and unseen scores close.

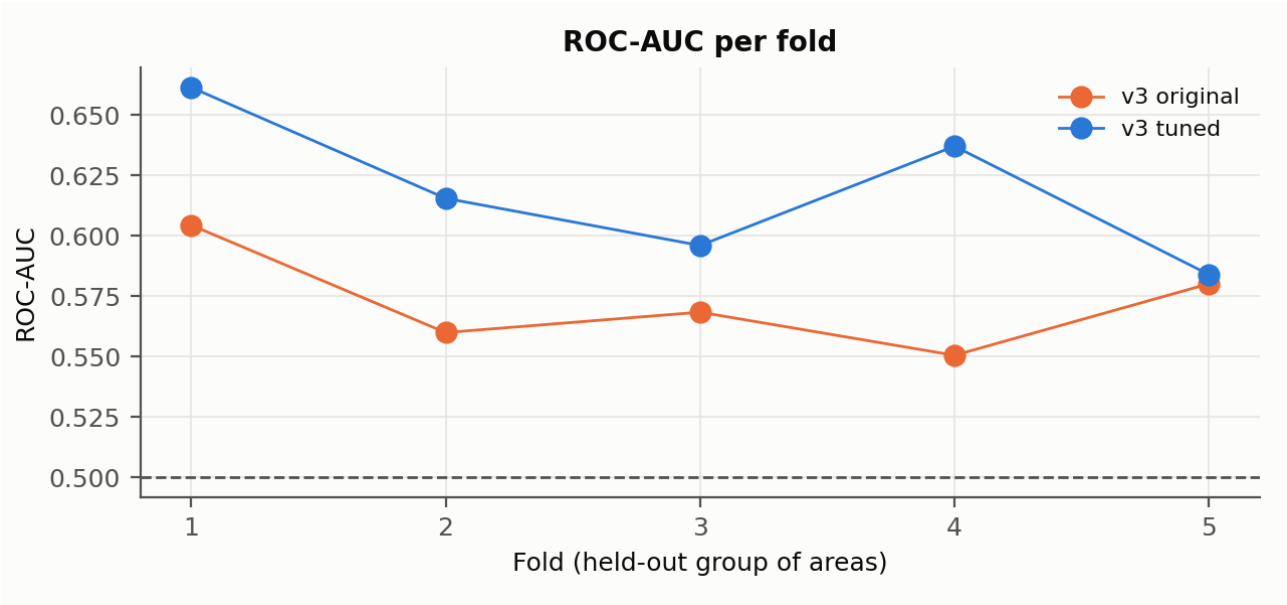


Figure 11. ROC-AUC per held-out fold.

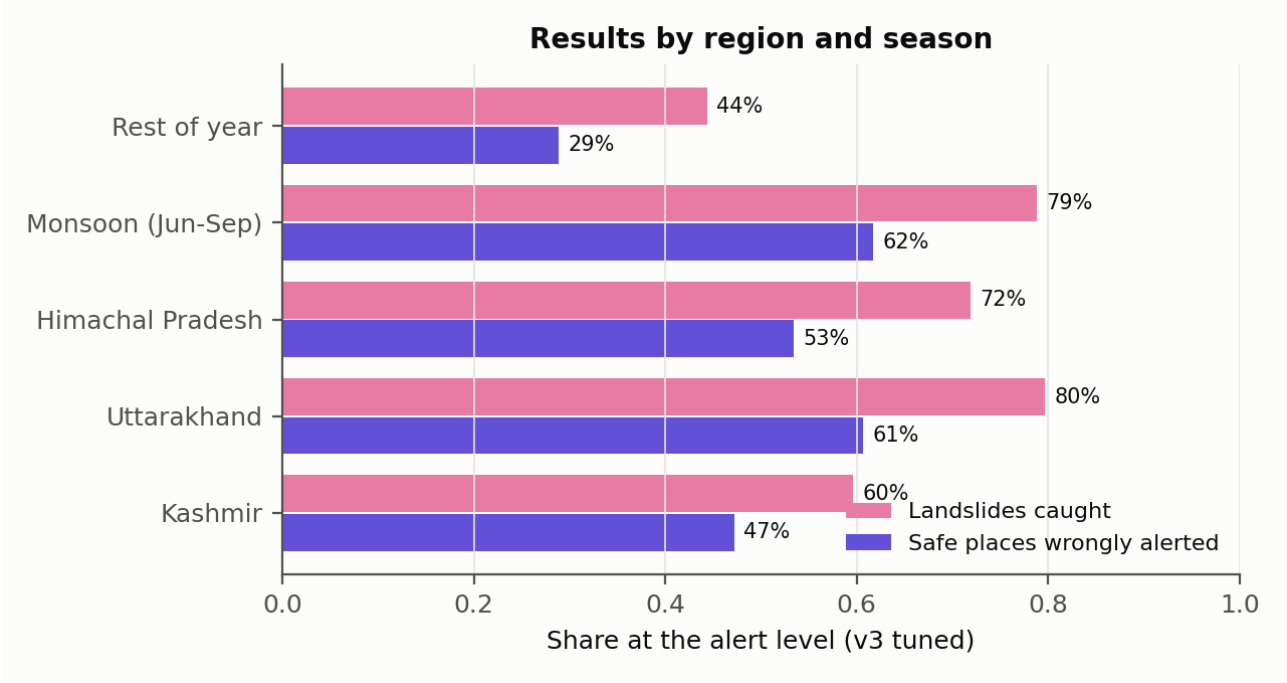


Figure 12. Recall and false-alarm rate by region and season at the alert level (groups with 50+ rows).

Results by group

Group	Rows	Landslides	Recall	False alarm rate	ROC-AUC
Kashmir	593	203	60%	47%	0.594
Uttarakhand	543	187	80%	61%	0.628
Himachal Pradesh	447	153	72%	53%	0.624
Haryana	12	4	75%	50%	0.656
NCT	12	4	75%	25%	0.719
Uttar Pradesh	3	1	0%	0%	1.000
Monsoon (Jun-Sep)	1194	410	79%	62%	0.625
Rest of year	416	142	44%	29%	0.593

Groups with fewer than 50 rows (plains states) are too small to judge; the app's plains rule handles them.

4. Discussion and limitations

What works: the model separates landslide days from normal days at the same location, and its scores rise with rain as physics requires. Tuning removed memorisation without changing the 20 inputs.
What does not: neighbouring slopes on the same day are barely separable (rain is measured in ~10 km squares and catalogue locations can be off by a few km), so false alarms stay high at useful recall.
Generalisation: results vary by region and season (Figure 12). The landslide catalogue is small (552 events, 2007-2016) and news-based, so remote and recent events are under-represented, and evaluation shares that bias.
Use: the outputs should be presented as risk warnings alongside official advisories, never as a statement that a landslide will occur.

5. Reproducibility

Seed 0 for fold splits, bootstrap seed 42, 1000 resamples. Tuned settings: {"params": {"max_depth": 4, "n_estimators": 100, "learning_rate": 0.02, "min_child_weight": 20, "subsample": 0.6, "colsample_bytree": 0.5, "reg_lambda": 10, "gamma": 0.5}, "monotone_rain": true, "ignore_unstable": true}. Re-run: *py -3.11 ai-module/v3/research_evaluation_v3.py*. All numbers are also in evaluation_report/metrics.json and tables/*.csv.